

Election Forensic Analytics For RCOUNT Packages: A Report Contract For Investigative, Non-Certifying Anomaly Analysis

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Draft — June 28, 2026

Abstract

Election forensic analytics look for unusual patterns in reported results: outlying precinct swings, turnout and vote-share residuals, batch and scanner discontinuities, suspicious digit distributions, spatial outliers, abrupt change points, and clustered audit discrepancies. These methods are valuable for triage, but they cannot certify an outcome or prove fraud. This paper defines the W-series, a family adjacent to the RCOUNT count-verification track but with a deliberately different claim boundary. It surveys the analytic families that hash-bound RCOUNT packages can feed, specifies a common forensic report contract every W-series method must satisfy — source package hashes, feature set and baseline population, model identity, unit-level scores, explicit caveats, and a plain investigative-not-certifying statement — and fixes the rule that anomaly scores must never masquerade as certification evidence. It is a framework and contract paper: the analytic methods are established, the report contract is specified here, and implementation placement (RCOUNT audit reports versus a later forensics crate) is left as an explicit open decision.

1 Introduction

Election forensic analytics ask whether reported results contain unusual patterns: precinct swings that do not fit their history, turnout or vote-share residuals far from comparable units, batch and scanner effects, leading- or last-digit distributions that look improbable, neighbors that diverge unexpectedly, abrupt change points across time or batches, and audit discrepancies that cluster by machine or source. These analytics are useful for prioritizing where humans should look. They do not, by themselves, certify an outcome or prove that anything went wrong.

That distinction is the reason for a separate W-series. The V-series verifies canvass and audit claims with explicit evidence contracts: a passing V-series transcript means a declared count or audit claim was replayed from public inputs. A W-series report makes no such claim. It produces an investigation queue — a ranked set of units, batches, scanners, or contests that merit more review — and it must say so in plain language on its face.

This paper leads with that boundary and then builds the shared scaffolding the rest of the track depends on. It surveys the analytic families that clean RCOUNT inputs can feed, specifies a single forensic report contract every method must satisfy, and lays out the fixture pattern that lets a report be reproduced from hash-bound inputs. It is a framework paper. The analytic methods it names are established in the forensics literature; the contribution here is the RCOUNT-grounded report contract and the strict separation that keeps anomaly scores from being read as certification evidence.

2 Analytic Families

RCOUNT packages provide clean, hash-bound inputs: source hashes, reporting-unit lineage, normalized summaries, batch manifests, cast vote records, and audit results. The W-series computes reproducible anomaly reports over those records. The pipeline is uniform across families:

certified/public results $\rightarrow$ normalized features $\rightarrow$ anomaly score $\rightarrow$ investigation queue.

The families differ in the question they ask and the RCOUNT records they consume.

Family	What it asks	RCOUNT input
Turnout residuals	Is turnout unusual after accounting for history and demographics?	summaries, precinct lineage
Vote-share residuals	Are candidate shares unusual relative to comparable units?	summaries, CVRs
Batch/scanner effects	Do batches or scanners show systematic deviations?	batch manifests, CVRs
Digit tests	Do reported numbers show suspicious digit patterns?	summaries
Benford-style checks	Are leading-digit distributions unusual where applicable?	large count tables
Spatial outliers	Are neighboring units unexpectedly different?	reporting units, geography
Change-point tests	Did patterns shift abruptly across time or batches?	status events, batches
Audit discrepancy clustering	Do discrepancies cluster by source, machine, or batch?	audit observations

Each family is the subject of a later W-series paper. What unifies them is not a single statistic but a shared input discipline and a shared output contract: the features are derived from hash-bound RCOUNT records, and the result is an anomaly report, never a pass or fail.

3 Report Contract

Every W-series method emits the same report shape so that any analytic can be reproduced from public inputs and read with its caveats attached. The per-record transcript fields are:

Field	Meaning
<code>method_id</code>	Registered forensic analytic.
<code>feature_set</code>	Variables used by the model.
<code>baseline_population</code>	Comparison units and time window.
<code>model_id</code>	Residual, digit test, spatial model, change-point, etc.
<code>unit_id</code>	Reporting unit, batch, scanner, or contest scored.
<code>score</code>	Anomaly score or test statistic.
<code>p_value_or_rank</code>	Optional statistical calibration.
<code>investigation_note</code>	Machine-readable caveat for the scored unit.

A report is also bound to its inputs. Every W-series report must carry the source package hashes it was computed from, the feature set and baseline population that define the comparison, the model and method identity, the unit-level scores, the caveats and false-positive boundary, and a plain statement that the output is investigative and not certifying. Binding the report to source package hashes is what makes it reproducible: a second analyst with the same RCOUNT package and the same registered method should recompute the same scores.

The report contract is specified here; its implementation placement is an open decision. The analytics could live as RCOUNT audit-style reports beside the existing transcripts, or in a later dedicated forensics crate. Either way, the W-series transcript must remain a distinct artifact from V-series certification evidence, so that an anomaly score is never stored or surfaced as a pass or fail.

4 Fixture Pattern

The W-series is validated by a small, deliberately shaped fixture family rather than by claims about any particular election. Four fixture roles recur across methods and define what a forensic report must and must not say.

Fixture	Role
No-anomaly	A synthetic package with clean, well-behaved features. A correct method returns low scores and an empty or low-priority queue.
Injected outlier	A package with one planted anomalous unit (for example an outlying precinct). The method must surface that unit near the top of the queue.
Batch/scanner effect	A package with a systematic batch- or scanner-level deviation. The method must attribute the effect to the right grouping rather than to individual units.
False-positive boundary	A package whose pattern is unusual but has an innocent explanation. The report must show why a high score is not proof, exercising the investigation-note and caveat fields.

The fixtures separate two things a forensic method must get right: detecting a genuinely planted anomaly, and refusing to escalate an unusual-but-explainable pattern into a fraud

claim. The false-positive boundary fixture is the most important of the four, because it is where the report contract earns its keep: the same machinery that produces a high score must also produce the caveat that keeps the score from being read as certification evidence.

These fixtures describe the intended evidence pattern for the track. As with the report contract, building them is downstream work that depends on normalized RCOUNT inputs; this paper fixes their roles so that later method papers and any implementation share one definition of success.

5 Boundaries

Every W-series page must lead with the same boundary, and this overview fixes it. Forensic analytics can prioritize investigation. They cannot prove fraud, certify correctness, or replace risk-limiting audits. A high anomaly score is a reason to look more closely, not a finding. The investigation-note and false-positive boundary fields exist precisely so that a score is never circulated without its caveat.

The separation from the V-series is structural, not just rhetorical. V-series algorithms verify audit and canvass claims with explicit evidence contracts; their transcripts are designed to support certification-style claims when their assumptions hold. W-series analytics explore patterns that may motivate more review; their transcripts are designed to support triage. RCOUNT must keep the two transcript kinds distinct so that an anomaly report can never be presented, stored, or audited as if it were certification evidence.

Two further boundaries follow. First, forensic analytics need clean public data; their value is bounded by the quality and completeness of the normalized RCOUNT inputs, and a sparse or biased baseline population can manufacture apparent anomalies. Second, the analytics are descriptive: they identify patterns that need explanation, and the explanation — benign or otherwise — is an external, human judgment that the W-series deliberately does not make.

6 Conclusion

The W-series gives RCOUNT a place to put investigative analytics without letting them drift into certification. This overview fixes the three things the rest of the track shares: the analytic families that hash-bound RCOUNT packages can feed, a single forensic report contract that binds every method to its source hashes and carries its caveats, and a fixture pattern that rewards detecting planted anomalies while refusing to escalate explainable ones.

The methods themselves — turnout and vote-share residuals, batch and scanner effects, digit and Benford tests, spatial outliers, change-point tests, and audit-discrepancy clustering — are the subjects of the W-series method papers that build on this contract. What this paper establishes is the discipline that makes them safe to publish: anomaly scores are an investigation queue, bound to public inputs and stated as investigative, never as proof and never as a certification pass or fail.

References

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